LESSON FIVE

CREATING FOLDERS, ADDING FILES, WRITING README AND USING MARKDOWN

5.1 Creating Folders Inside a GitHub Repository

I learned that folders are used to organize files in a project. A well-organized repository makes it easier for developers to understand and maintain their work.

For example, when developing a website, I can create different folders for CSS files, JavaScript files and images.

A typical website project may have the following structure:

```
my-project/
│
├── index.html
├── about.html
├── README.md
│
├── css/
│   └── style.css
│
├── js/
│   └── script.js
│
└── images/
    └── logo.png
```

I learned that I can create folders locally on my computer and push them to GitHub with their files.

I can also create folders directly on GitHub by selecting the option to create a new file and including the folder name followed by a forward slash.

For example:

```
images/logo.png
```

This creates a file named logo.png inside an images folder.

Git does not track empty folders by themselves. A folder normally needs to contain at least one file to appear in a repository.

5.2 Adding Files to a Repository

I learned that files can be added to a GitHub repository in different ways.

Method One: Using the GitHub Website

1. Open my repository on GitHub.

2. Select Add file.
3. Choose Create new file or Upload files.
4. Enter the file name or select files from my computer.
5. Add a commit message.
6. Save the changes.

Method Two: Using the Terminal

I can add files to my local repository using:

```
git add .
```

Then I can save the changes:

```
git commit -m "Added project files"
```

Finally, I can upload the changes:

```
git push
```

5.3 Meaning of README.md

I learned that README.md is a special file commonly used to provide information about a project. It is usually displayed automatically on the main page of a GitHub repository.

The README file helps visitors understand what a project does, the technologies used and how to install or use it.

5.4 Important Sections of a README File

A good README file may contain the following sections:

1. **Project Title:** The name of the project.
2. **Description:** A short explanation of what the project does.
3. **Features:** The important functions of the project.
4. **Technologies Used:** The programming languages and tools used.
5. **Installation:** Instructions for setting up the project.
6. **Usage:** How to use the application.
7. **Screenshots:** Images showing the application interface.
8. **Contributors:** Names of people who worked on the project.
9. **License:** Information about the conditions for using the project.

5.5 Meaning of Markdown

I learned that Markdown is a lightweight markup language used to format text. GitHub uses Markdown in README files, documentation and other areas.

Markdown allows me to create headings, paragraphs, lists, links, images and code blocks without writing complicated HTML.

5.6 Common Markdown Syntax

1. Headings

```
# Main Heading
## Second Heading
### Third Heading
```

2. Bold Text

```
**This text is bold**
```

3. Italic Text

```
*This text is italic*
```

4. Unordered List

```
- HTML
- CSS
- JavaScript
- PHP
```

5. Ordered List

```
1. Create a repository
2. Add files
3. Commit changes
4. Push to GitHub
```

6. Links

```
[Visit Google](https://www.google.com)
```

7. Inline Code

```
Use `git status` to check the repository.
```

8. Code Block

Three backticks can be used to display a block of code:

````
```php
<?php
echo "Hello World";
?>
```
````

5.7 Displaying Images in README.md

I learned that Markdown allows me to display images inside a README file. This is useful when I want to show screenshots of my website, application or project interface.

The general syntax is:

```
![Alternative Text](image-url)
```

For example:

```
![Project Logo](images/logo.png)
```

This displays the logo.png image stored inside the images folder.

I can also display an image hosted online by using its image URL.

The alternative text describes the image and can help readers understand what the image represents if it cannot be displayed.

5.8 Importance of README and Markdown

I learned that README and Markdown are important because they:

1. Explain the purpose of a project.
2. Make repositories easier to understand.
3. Help developers document their work.
4. Allow screenshots and other images to be displayed.
5. Make project documentation look organized.
6. Help other developers install and use a project.

Conclusion

From this lesson, I understood how to organize project files into folders, add files to GitHub, write a README file and use Markdown syntax to format information. I also learned how to display project images inside a README file.